

flag_trait_outliers.R

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Contents

flag_trait_outliers 1

Index 3

flag_trait_outliers *Flag Trait Outliers Within Groups*

Description

Adds ‘n_group_records’ and ‘is_trait_outlier’ columns to grouped trait records using a symmetric median-IQR fence. The caller is responsible for grouping ‘data_trait_records’ with ‘dplyr::group_by()’ before calling this function. The returned data frame is always ungrouped.

Usage

```
flag_trait_outliers(  
  data_trait_records,  
  trait_value_column = "trait_value",  
  iqr_multiplier,  
  minimum_group_size = NULL  
)
```

Arguments

data_trait_records
A grouped data frame (e.g. created by ‘dplyr::group_by()’). Must contain the column named by ‘trait_value_column’.

trait_value_column
Character scalar. Name of the numeric column to test. Default: “trait_value”.

iqr_multiplier
Positive numeric scalar. IQR fence multiplier. A value is flagged when ‘|value - group_median| > iqr_multiplier * group_IQR’.

minimum_group_size
Optional positive integer scalar. When supplied, ‘is_trait_outlier’ is set to ‘FALSE’ for groups with fewer than ‘minimum_group_size’ records or with an IQR of zero. Default: ‘NULL’ (no group-size guard applied).

Details

The IQR fence is `'[group_median - iqr_multiplier * group_IQR, group_median + iqr_multiplier * group_IQR]'`. Values outside this range receive `'is_trait_outlier = TRUE'`.

When `'minimum_group_size'` is not `'NULL'`, an additional guard is applied: groups with fewer than `'minimum_group_size'` records or a `'group_IQR'` of zero always receive `'is_trait_outlier = FALSE'`.

Designed for use in pipes where the caller sets up grouping: `“r data_traits |> dplyr::group_by(trait_domain_name) |> flag_trait_outliers(iqr_multiplier = 3) “`

Value

An ungrouped data frame with the same rows and columns as `'data_trait_records'` plus two additional columns:

`'n_group_records'` Integer. Number of records in the group.

`'is_trait_outlier'` Logical. `'TRUE'` when the record's value falls outside the IQR fence; `'FALSE'` otherwise. When `'minimum_group_size'` is supplied, also `'FALSE'` for groups that are too small or have zero IQR.

See Also

`[filter_trait_outliers()]`, `[write_trait_quality_control_report()]`

Index

`flag_trait_outliers`, [1](#)